HAMILTONIAN FIXED POINTS BELOW THE STABLE MORSE NUMBER IN DIMENSION TEN, AND AN OBSTRUCTION IN DIMENSION SIX

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Abstract. We construct a simply connected closed Kähler manifold M of real dimension ten and a Hamiltonian diffeomorphism of M with only nondegenerate fixed points whose number is $\lambda(M) - 4 = \mathrm{SM}(M) - 4$, where λ is the integral Morse count and SM is the stable Morse number. The count equals the cyclic integral Floer bound of Bai–Xu, which is therefore sharp already in complex dimension five. The new ingredients are a simply connected threefold F with Brauer-type 2-torsion, obtained as a resolved $\mathbf{Z}/5$-quotient of an invariant Artin–Mumford double solid, a divisor $S \subset F$ with $\pi_1(S) = \mathbf{Z}/5$, and a blow-up along a center that meets the fixed locus of the Hamiltonian involution transversally, so that no new fixed components appear. In the other direction, the Bai–Xu bound excludes simply connected counterexamples in real dimension six, and in real dimension eight every construction of the type of Proposition 2.6 (a finite-order element of a Hamiltonian torus action plus a small invariant perturbation) yields at least $\lambda(M)$ fixed points. Among simply connected closed symplectic manifolds of real dimension at least six, the minimal dimension of a counterexample to the stable Morse bound is therefore 8 or 10.

Status. Internal draft. Theorem numbers cited from [OpenAI22] refer to its October 5, 2026 build. Every point that has not been checked against the primary literature is marked [**TODO:** . . .]. Nothing in this document has been circulated.

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Date: October 9, 2026 (internal; not for distribution).

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1. Introduction

1.1. **Stable Morse number and Hamiltonian fixed points.** Let W be a connected closed smooth manifold. Its *stable Morse number* $\mathrm{SM}(W)$ is the minimum of $\#\operatorname{Crit}(\Phi)$ over $k \geq 0$ and Morse functions $\Phi\colon W \times \mathbf{R}^k \to \mathbf{R}$ that coincide with a nondegenerate quadratic form in the $\mathbf{R}^k$-variables outside a compact set; the case $k = 0$ is the ordinary Morse number $\mathrm{Morse}(W) \geq \mathrm{SM}(W)$. For a closed symplectic manifold (M, ω) and a one-periodic Hamiltonian H all of whose fixed points are nondegenerate, the *stable Morse formulation* of the Arnold problem asks whether

$$\#\operatorname{Fix}_0(\phi_H^1; H) \;\geq\; \mathrm{SM}(M), \tag{1.1}$$

where Fix_0 denotes fixed points with contractible orbits; see [Gol20, Conjecture 1.2]. Positive results are known under asphericity hypotheses [DG17]. Counterexamples on simply connected Kähler manifolds were found in [OpenAIStable; OpenAI22; OpenAICompanion], the smallest previously recorded fixed dimension being real dimension 22 [OpenAI22]; a non-simply-connected counterexample to the ordinary Morse-number bound in real dimension 12 is given in [OpenAISept23].

For a closed manifold W put

$$b(W) = \sum_i \operatorname{rank} H_i(W; \mathbf{Z}), \qquad t_i(W) = d(\operatorname{Tor} H_i(W; \mathbf{Z})), \qquad \lambda(W) = b(W) + 2\sum_i t_i(W),$$

where d is the minimal number of generators. Then $\mathrm{SM}(W) \geq \lambda(W)$ for every closed W [OpenAI22, Prop. 2.2], with equality for simply connected W of dimension ≥ 6 by Smale's theorem (§2.2).

1.2. **Main results.** The following theorem lowers the dimension of a simply connected counterexample from 22 to 10.

Theorem A. *There exist a simply connected closed Kähler manifold (M, ω) of complex dimension 5 and a smooth one-periodic Hamiltonian H on M such that every fixed point of ϕ_H^1 is nondegenerate and*

$$\#\operatorname{Fix}(\phi_H^1) = \#\operatorname{Fix}_0(\phi_H^1; H) = \lambda(M) - 4 = \mathrm{SM}(M) - 4 = \mathrm{Morse}(M) - 4.$$

Moreover the minimal Chern number of M is 1 and $\#\operatorname{Fix}(\phi_H^1)$ equals the cyclic integral Floer bound $\lambda_1(M)$ of Bai–Xu (Definition 2.1); this bound is therefore attained in complex dimension 5.

The gap 4 does not depend on any free Betti number. For the resolution chosen in §4 we obtain $\mathrm{SM}(M) = 524$ and $\#\operatorname{Fix} = 520$ (Remark 5.5).

Theorem B. *Let (M,ω) be a closed symplectic manifold with a Hamiltonian action of a torus T, and let $g \in T$ have finite order. Suppose that every connected component $C_1,\dots,C_r$ of $\mathrm{Fix}(g)$ has real dimension at most 6 (this holds whenever $\dim_{\mathbf{R}} M = 8$ and g does not act trivially). Then*

$$\sum_{j=1}^{r} \lambda(C_j) \;\geq\; \lambda(M).$$

Consequently, if $\dim_{\mathbf{R}} M = 8$, every Hamiltonian diffeomorphism $\psi^{\epsilon} \circ g$ produced by Proposition 2.6 has at least $\lambda(M)$ nondegenerate fixed points, and at least $\mathrm{SM}(M)$ if M is simply connected.

Theorem B is complemented by necessary conditions valid for every simply connected eight-dimensional counterexample (Proposition 6.1): the minimal Chern number must be 1, and the dominant primes of $\mathrm{Tor}\, H_2(M;\mathbf{Z})$ and $\mathrm{Tor}\, H_3(M;\mathbf{Z})$ must differ. All previously known simply connected constructions [OpenAIStable; OpenAI22; OpenAICompanion] and the construction of Theorem A are of the type excluded by Theorem B in dimension 8. In dimension 6 the Bai–Xu bound already excludes simply connected counterexamples (Proposition 3.1); real dimensions 2 and 4 are discussed in §3. In summary:

Corollary 1.1. *Among simply connected closed symplectic manifolds of real dimension at least 6, the minimal dimension of a counterexample to (1.1) is 8 or 10.*

real dim	status	reference
2	no counterexample	§3
4	simply connected: open, reduces to SM vs. λ	§3
6	no simply connected counterexample	Prop. 3.1
8	open; excluded within the finite-order/torus mechanism	Theorem B, Prop. 6.1
10	simply connected Kähler counterexample, Floer bound sharp	Theorem A
12	non-simply-connected counterexample to the Morse-number bound	[OpenAISept23]
12, 16	simply connected Kähler counterexamples	Remarks 7.1, 7.2
22	simply connected Kähler family, unbounded deficit	[OpenAI22]

TABLE 1. Current status by dimension (nondegenerate Hamiltonians).

1.3. **Previous work versus our contributions.** *Previous work.* The mechanism of [OpenAIStable; OpenAI22; OpenAICompanion] takes $M = X \times Y$, where X is a simply connected Kähler manifold with torsion at two primes and Y is $\mathbf{CP}^1 \times \mathbf{CP}^1$ blown up at four torus-fixed points; the half-turn g_Y of a Hamiltonian circle action on Y fixes four lines, and a small invariant perturbation of $\mathrm{id}_X \times g_Y$ has $4\lambda(X \times \mathbf{CP}^1)$ fixed points. Because d takes the *maximum* over primes in each degree, convolution of the torsion profile of X with the free profiles $(1,1)$ of $\mathbf{CP}^1$ and $(1,6,1)$ of Y can make $\lambda(X \times Y) > 4\lambda(X \times \mathbf{CP}^1)$. The torsion of X comes from blow-ups of a free ambient space along bielliptic surfaces and their products, and the resulting examples live in real dimension 22 [OpenAI22] (and 3332 in [OpenAICompanion], where the cyclic Floer bound is attained). Independently, [OpenAISept23, Thm. 1.1] gives a nondegenerate counterexample to the *ordinary* Morse-number bound on a closed symplectic manifold of real dimension 12 with fundamental group A_5^k (not simply connected), with $\#\,\mathrm{Fix} = \chi(M) + 128 < \mathrm{Morse}(M)$; there the effect comes from π_1, not from torsion. A degenerate six-dimensional example on a quadric with three fixed points, for the critical-point count, is in [OpenAIDeg]. The stable Morse bound is formulated in [Gol20]; Bai–Xu [BX22; BX26] prove the cyclic integral lower bound. Ma's preprint [Ma13, Thm. 1.5] asserts that the minimal number of fixed points equals the Morse number; see §7.2.

Where the previous method fails. The product mechanism cannot work below real dimension 12 (Proposition 6.3): for a simply connected X of real dimension ≤ 6 the torsion of

X is a single group placed in degrees $2, 3$, and no exchange of dominant prime can occur. Moreover blow-ups of a free ambient space never create torsion in degree 2, which is forced in dimension 8 by Proposition 6.1.

Our contributions.

(1) A new torsion source: a simply connected threefold F with $\operatorname{Tor} H_2(F) = \operatorname{Tor} H_3(F) = \mathbf{Z}/2$ and *no* 5-torsion, containing a smooth divisor S with $\pi_1(S) = \mathbf{Z}/5$ (§4). The proof that the 5-torsion vanishes uses a permutation-module structure of H_2 of an Artin–Mumford double solid and Lefschetz duality.

(2) A non-product construction: the blow-up of $F \times Y$ along $W = S \times (L_0 \sqcup L_2)$, where $L_0, L_2 \subset Y$ are invariant lines meeting the fixed curves transversally. The transverse center lemma (Lemma 5.1) shows that this blow-up adds torsion to the fixed components only in a controlled way and creates no new fixed components (§5). This gives Theorem A: the first simply connected example below the stable Morse number in real dimension < 22, and a counterexample to the Morse-number assertion of [Ma13] on a simply connected Kähler manifold of real dimension 10.

(3) In dimension 8, necessary conditions together with the unconditional mechanism obstruction Theorem B, based on perfectness of moment maps over every prime field (Lemma 6.4, a specialization of the Borel–Smith inequality; we do not claim novelty for it).

(4) A heuristic explanation of why the transverse-center construction needs real dimension at least 10 (§6.4), and a machine classification of admissible torsion configurations corroborating Theorem B (Appendix A).

The low-dimensional statements of §3 are straightforward consequences of Bai–Xu and duality. The Kähler examples in dimensions 12 and 16 recorded in §7.1 are rearrangements of the same ideas and are not claimed as separate results.

1.4. **Conventions.** Homology has integer coefficients unless indicated. For a finite abelian group A and a prime p, write $d_p(A) = \dim_{\mathbf{F}_p}(A \otimes \mathbf{F}_p)$; then $d(A) = \max_p d_p(A)$. For a space X set $t_i^p(X) = d_p(\operatorname{Tor} H_i(X))$ and $r_i(X) = \operatorname{rank} H_i(X)$. Homology shifted by $2k$ is written $H(\cdot)[2k]$.

2. Torsion counts and fixed-point bounds

2.1. **Degreewise and cyclic generator counts.**

Definition 2.1. For a closed manifold W and $N \in \{1, 2, \dots\} \cup \{\infty\}$ put

$$\lambda_N(W) = b(W) + 2 \sum_{c \in \mathbf{Z}/2N} d\Big(\bigoplus_{i \equiv c \ (2N)} \operatorname{Tor} H_i(W) \Big),$$

with $\mathbf{Z}/2\infty = \mathbf{Z}$, so that $\lambda_\infty = \lambda$.

Since $d(A \oplus B) = \max_p(d_p(A) + d_p(B)) \le d(A) + d(B)$, one has $\lambda_N \le \lambda$, with equality as soon as each residue class mod $2N$ contains at most one degree carrying torsion. For finite groups A, B,

$$d(A) + d(B) - d(A \oplus B) > 0 \iff \operatorname{argmax}_p d_p(A) \cap \operatorname{argmax}_p d_p(B) = \varnothing \text{ and } A, B \neq 0. \tag{2.1}$$

We say that A and B have *different dominant primes* in this case.

2.2. **Morse-theoretic bounds.**

Proposition 2.2 ([OpenAI22, Prop. 2.2])**.** *For every connected closed smooth manifold W, $\mathrm{SM}(W) \ge \lambda(W)$; in particular every Morse function on W has at least $\lambda(W)$ critical points.*

Theorem 2.3 (Smale)**.** *Let W be a simply connected closed smooth manifold of dimension $D \ge 6$. Then W admits a Morse function with exactly $r_i(W) + t_i(W) + t_{i-1}(W)$ critical points of index i; hence $\mathrm{SM}(W) = \mathrm{Morse}(W) = \lambda(W)$.*

For $D \geq 9$ this is [OpenAI22, Thm. 2.4] with the handle proof of its appendix, based on [Sma63]. For all $D \geq 6$ it is [Sma62, Thm. 6.1].[1]

2.3. **The cyclic integral Floer bound.** Let N_M be the minimal Chern number of (M, ω): the positive generator of $c_1(TM)(\pi_2(M)) \subset \mathbf{Z}$, or ∞ if this subgroup is zero (Bai–Xu write $N = 0$ in this case).

Theorem 2.4 (Bai–Xu [BX22, Thm. A], see also [BX26, Cor. 1.2])**.** *Let (M, ω) be any closed symplectic manifold and H a one-periodic Hamiltonian with only nondegenerate fixed points. Then*

$$\#\operatorname{Fix}_0(\phi_H^1; H) \;\geq\; \operatorname{rank} H_*(M; \mathbf{Q}) + 2 \sum_{i \in \mathbf{Z}/2N_M} \tau_i^{(2N_M)}(M) = \lambda_{N_M}(M),$$

where $\tau_i^{(2N)}(M)$ is the number of nontrivial invariant factors of the torsion of $\bigoplus_{k \equiv i \ (2N)} H_k(M; \mathbf{Z})$, that is, its minimal number of generators.

Remark 2.5. Rezchikov [Rez22, Thm. 1] phrases an integral lower bound as the minimal rank of a cochain complex of free $\mathbf{Z}$-modules with cohomology $H^*(M; \mathbf{Z})$. His proof works with the $2N$-periodized Morse complex [Rez22, §2.1], so the bound obtained is the cyclic one, consistent with Theorem 2.4. Read with a $\mathbf{Z}$-grading, the statement would give $\#\operatorname{Fix} \geq \lambda(M)$, which Theorem A contradicts.

2.4. **Perturbation of finite-order Hamiltonian symmetries.** The following is [OpenAI22, Prop. 4.2] (Prop. dyn:involution there) with the involution replaced by an element of finite order.

Proposition 2.6. *Let (P, ω) be a closed symplectic manifold and g a Hamiltonian diffeomorphism with $g^k = \mathrm{id}$ for some $k \geq 1$. Let $\operatorname{Fix}(g) = C_1 \sqcup \cdots \sqcup C_r$ and let $h_j \colon C_j \to \mathbf{R}$ be Morse functions. There is a smooth g-invariant $f \colon P \to \mathbf{R}$ with $f|_{C_j} = h_j$ such that for all sufficiently small $\epsilon > 0$ the Hamiltonian flow ψ^s of f satisfies*

$$\operatorname{Fix}(\psi^\epsilon \circ g) = \bigsqcup_j \operatorname{Crit}(h_j),$$

every such fixed point is nondegenerate, and $\psi^\epsilon \circ g$ is the time-one map of a smooth one-periodic Hamiltonian. If P is simply connected, all fixed-point orbits are contractible.

Proof. We indicate the changes to the proof of [OpenAI22, Prop. 4.2]. Average a metric over $\langle g \rangle$; the fixed set is a union of closed submanifolds with $T_x C_j = E_x^+ = \ker(d_x g - I)$. Let $E_x^{\neq}$ be the sum of the (real parts of the) generalized eigenspaces of $d_x g$ for eigenvalues $\neq 1$; $d_x g$ is semisimple of finite order, so $T_x P = E_x^+ \oplus E_x^{\neq}$ and $E_x^{\neq} = \operatorname{im}(d_x g - I)$. If $u \in E_x^+$ and $v = d_x g(w) - w$, then $\omega(u, v) = \omega(d_x g\, u, d_x g\, w) - \omega(u, w) = 0$, so the summands are ω-orthogonal and C_j is symplectic. Let $f = \frac{1}{k} \sum_{m=0}^{k-1} \widetilde{f} \circ g^m$ for an extension $\widetilde{f}$ of the h_j. Then f is g-invariant, $d_x f$ vanishes on $\operatorname{im}(d_x g - I) = E_x^{\neq}$ for $x \in \operatorname{Fix}(g)$, hence X_f is tangent to C_j and equals X_{h_j} there, and g commutes with the flow ψ^s of X_f. Put $\Phi = \psi^\epsilon \circ g$; then $\Phi^k = \psi^{k\epsilon}$. If $\Phi(x) = x$ and $k\epsilon$ is smaller than the constant of [OpenAI22, Lemma 4.1] (short-period lemma), then $X_f(x) = 0$, so $\psi^\epsilon(x) = x$ and $g(x) = x$, and $x \in \operatorname{Crit}(h_j)$. At such x, with $A = d_x X_f$ preserving E_x^+ and $E_x^{\neq}$, the tangent block of $d_x \Phi - I$ is $\exp(\epsilon A^+) - I = \epsilon A^+ + O(\epsilon^2)$ with A^+ invertible by the Morse condition, and the normal block is $d_x g|_{E^{\neq}} \exp(\epsilon A^{\neq}) - I \to d_x g|_{E^{\neq}} - I$, which is invertible. The concatenation argument producing a one-periodic Hamiltonian is unchanged. □

[1] The torsion-inclusive statement of [Sma62, Thm. 6.1] for $n \geq 6$ was confirmed through secondary sources (Encyclopedia of Mathematics, "Morse inequalities"; arXiv:2607.20259); the original wording was not consulted.

2.5. **Integral blow-up formula.**

Proposition 2.7 ([DL09, Thm. 4.1], [OpenAI22, Prop. 3.3])**.** *Let A be a compact connected Kähler manifold and $Z \subset A$ a closed connected complex submanifold of complex codimension $c \geq 2$. Then $\mathrm{Bl}_Z A$ is compact Kähler, simply connected if A is, and*

$$H_i(\mathrm{Bl}_Z A; \mathbf{Z}) \cong H_i(A; \mathbf{Z}) \oplus \bigoplus_{j=1}^{c-1} H_{i-2j}(Z; \mathbf{Z}).$$

3. Low dimensions

The statements of this section are direct consequences of Theorem 2.4 and Poincaré duality; we record them to delimit Corollary 1.1.

Real dimension 2. For a closed oriented surface Σ_g, SM = Morse = $\lambda = 2g + 2$ (Proposition 2.2 and the standard Morse function). If $g = 0$, $N = 2$ and $\lambda_2(S^2) = 2$; if $g \geq 1$, $\pi_2 = 0$, $N = \infty$ and $\lambda_\infty = \lambda$. Theorem 2.4 gives $\#\operatorname{Fix}_0 \geq 2g + 2 = \mathrm{SM}(\Sigma_g)$, so there is no counterexample.

Simply connected real dimension 4. The homology is torsion-free, so Theorem 2.4 gives $\#\operatorname{Fix} \geq b(M) = \lambda(M)$. A simply connected four-dimensional counterexample would therefore require $\mathrm{SM}(M) > \lambda(M)$. Smale's theorem does not apply in dimension 4, and we do not settle whether SM = λ for simply connected closed 4-manifolds here.

Proposition 3.1. *Let (M, ω) be a simply connected closed symplectic manifold of real dimension* 6*. Every one-periodic Hamiltonian with only nondegenerate fixed points satisfies $\#\operatorname{Fix}(\phi_H^1) \geq \lambda(M) = \mathrm{SM}(M)$.*

Proof. Write $T_i = \operatorname{Tor} H_i(M)$. By universal coefficients and Poincaré duality, $T_i \cong \operatorname{Tor} H^{i+1}(M) \cong T_{5-i}$. Since H_0 is free and $H_1 = 0$, we get $T_0 = T_1 = T_4 = T_5 = 0$, $T_6 = 0$, and $T_2 \cong T_3$. The two degrees 2 and 3 have different parities, hence lie in different residue classes modulo $2N$ for every $N \in \{1, 2, \ldots, \infty\}$. Therefore $\lambda_N(M) = \lambda(M)$ for every N. Theorem 2.4 gives $\#\operatorname{Fix} \geq \lambda(M)$, and Theorem 2.3 with $D = 6$ gives $\lambda(M) = \mathrm{SM}(M)$; contractibility is automatic. □

Remark 3.2. Proposition 3.1 concerns nondegenerate Hamiltonians. The six-dimensional example of [OpenAIDeg], a Hamiltonian diffeomorphism of the quadric Q^3 with three fixed points, has a degenerate fixed point and concerns the critical-point count; it does not conflict with Proposition 3.1.

4. An Artin–Mumford quotient threefold

4.1. **An invariant quartic symmetroid web.** Let $\zeta = e^{2\pi i/5}$ and let $G = \langle g \rangle \cong \mathbf{Z}/5$ act on $\mathbf{P}^3$ by

$$g \cdot [x_0 : x_1 : x_2 : x_3] = [x_0 : \zeta^4 x_1 : \zeta^3 x_2 : \zeta x_3].$$

Consider the web of quadrics given by the symmetric matrix of linear forms

$$A(x) = \begin{pmatrix} 6x_0 & 5x_1 & 4x_2 & 0 \\ 5x_1 & 8x_2 & 0 & 4x_3 \\ 4x_2 & 0 & 7x_3 & 6x_0 \\ 0 & 4x_3 & 6x_0 & 5x_1 \end{pmatrix},$$

and let $B = \{f = 0\}$, $f = \det A$:

$$f = 900x_0^2x_1^2 - 1728x_0^3x_2 - 640x_1x_2^3 - 875x_1^3x_3 + 720x_0x_1x_2x_3 + 256x_2^2x_3^2 - 672x_0x_3^3.$$

Every monomial of f has g-weight 3 mod 5, so $f(gx) = \zeta^3 f(x)$.

Lemma 4.1 (machine-verified, Appendix B)**.** (1) *The four matrices $A_0, \ldots, A_3$ (coefficients of $x_0, \ldots, x_3$) are linearly independent, the web has no base points, and no member has rank ≤ 1.*

(2) *f is irreducible over $\mathbf{Q}$, the singular scheme of B equals the rank-≤ 2 scheme of A, and it is reduced of length 10. Thus B has exactly ten singular points, all ordinary double points (A_1). Since $\mathrm{Sing}(B)$ is finite and a reducible surface is singular along a curve, f is absolutely irreducible. The only g-fixed points of $\mathbf{P}^3$ are the coordinate points, which are smooth points of B (item 3) and $|G| = 5$ is prime, so the nodes form two free G-orbits.*
(3) *The four coordinate points $p_0, \dots, p_3$ lie on B and are smooth points of B.*
(4) *(Beauville's condition (ii).) If a line of $\mathbf{P}^3$ is contained in the singular locus of a quadric of the web, it is not contained in any other quadric of the web. Equivalently, at each of the ten rank-2 points the three quadratic conditions on the web imposed by the kernel line are independent; this is checked exactly over $\overline{\mathbf{Q}}$.*
(5) *The ten nodes impose independent conditions on quadrics: the Hilbert function of $\mathbf{C}[x_0, \dots, x_3]/I_{\mathrm{nodes}}$ is $1, 4, 10, 10, \dots$, so $H^0(\mathbf{P}^3, I_{\mathrm{nodes}}(2)) = 0$.*

Let $\overline{X} \to \mathbf{P}^3$ be the double cover $w^2 = f$ (a hypersurface in $\mathbf{P}(1,1,1,1,2)$) and $X := \mathrm{AM}_5$ its blow-up at the ten nodes. The lift $\hat{g}\colon (x, w) \mapsto (gx, \zeta^4 w)$ has order 5 (note $\zeta^8 = \zeta^3$) and lifts to X; we denote it again by g.

Lemma 4.2. *g acts on X with exactly four fixed points, one over each p_k, each of type $\frac{1}{5}(1,1,4)$. All other points have trivial stabilizer.*

Proof. The weights $0, 4, 3, 1$ are pairwise distinct, so $\mathrm{Fix}(g|_{\mathbf{P}^3}) = \{p_0, \dots, p_3\}$; the nodes form free orbits, so no exceptional quadric contains a fixed point. Since $p_k \in B$, the fiber over p_k is the single point $w = 0$. At p_0 use affine coordinates $u_k = x_k/x_0$ of weights $4, 3, 1$. The linear part of $f(1, u)$ is a nonzero multiple of u_2 ($\partial f/\partial x_2(p_0) = -1728$); hence (w, u_1, u_3) are local coordinates on X, of weights $(4, 4, 1)$, i.e. type $\frac{1}{5}(1,1,4)$ after multiplying by 4. At p_1, p_2, p_3 the nonvanishing partials are $\partial_{x_3} f(p_1) = -875$, $\partial_{x_1} f(p_2) = -640$, $\partial_{x_0} f(p_3) = -672$, and the same computation gives weights $(1,1,4)$, $(3,2,3)$, $(2,3,2)$, all of type $\frac{1}{5}(1,1,4)$. As $|G|$ is prime, every non-fixed point has trivial stabilizer. □

4.2. **Topology of the invariant double solid.**

Proposition 4.3. *$X = \mathrm{AM}_5$ is simply connected and*

$$H_*(X; \mathbf{Z}) = (\mathbf{Z},\ 0,\ \mathbf{Z}^{11} \oplus \mathbf{Z}/2,\ \mathbf{Z}/2,\ \mathbf{Z}^{11},\ 0,\ \mathbf{Z}) \quad \textit{in degrees } 0, \dots, 6.$$

Proof. Betti numbers. By Lemma 4.1(5) the ten nodes impose independent conditions on quadrics, so the defect of the nodal quartic double solid $\overline{X}$ vanishes. Clemens' formulas for double solids [Cle83] [**TODO:** page/number of the defect formula] give, for a quartic double solid with $s = 10$ nodes and defect $\delta = 0$, $b_3 = 20 - 2s + 2\delta = 0$ for the small resolution and hence for X, and $b_2(X) = 1 + s + \delta = 11$. This agrees with the Euler characteristic: a quartic with ten nodes has $\chi = 24 - 10 = 14$, so $\chi(\overline{X}) = 2\chi(\mathbf{P}^3) - 14 = -6$, and replacing each node by a quadric adds 3, giving $\chi(X) = 24 = 2 + 2 \cdot 11$.

Nonzero 2-torsion. The web is base point free and satisfies Beauville's condition (ii) (Lemma 4.1(1),(4)). These are the hypotheses of [Bea15, §6.3], which then gives a nonzero 2-torsion class in $H^3(X; \mathbf{Z})$ (equivalently a nonzero element of the Brauer group), following [AM72].

Exactness and simple connectivity, by deformation. We do not know whether the computation of [AM72] that the torsion is exactly $\mathbf{Z}/2$ applies verbatim to every web satisfying (i) and (ii); we therefore deform to a general web. Let $\mathrm{Gr} = \mathrm{Gr}(4, \mathrm{Sym}^2 \mathbf{C}^4)$ (irreducible) and let $\mathcal{D} \subset \mathbf{P}^3 \times \mathrm{Gr}$ be the relative rank-≤ 2 scheme and $\mathcal{S} \supset \mathcal{D}$ the relative singular scheme of the symmetroids; $\mathcal{S}$ is proper over Gr. The partial derivatives of $\det A_\Lambda$ are combinations of 3×3 minors, so $\mathcal{D}_\Lambda \subset \mathcal{S}_\Lambda$. When $\mathcal{D}_\Lambda$ is finite it has the expected codimension 3 and length equal to the degree 10 of the corank-2 locus of symmetric 4×4 matrices [HT84] [**TODO:** confirm that the Harris–Tu degree computes scheme-theoretic length in the expected-codimension

case]. Hence length $\mathcal{S}_\Lambda \geq 10$ whenever it is finite. On the open locus over which $\mathcal{S}$ is quasi-finite, the fibre length is upper semicontinuous, and the étale locus of a finite flat family is open; hence

$$U_0 = \{\Lambda : \mathcal{S}_\Lambda \text{ is reduced of length } 10\}$$

is Zariski open in Gr. It contains the general web (the Artin–Mumford case) and our web (Lemma 4.1(2)); being a nonempty Zariski open subset of an irreducible variety, it is connected. Over U_0 the nodes form a finite étale multisection, and by the holomorphic Morse lemma with parameters the double solids are locally analytically of the form $w^2 = y_1^2+y_2^2+y_3^2$ times the base near the nodes. Blowing up the multisection therefore gives a smooth proper family, and by Ehresmann's theorem all members are diffeomorphic. For a general quartic symmetroid, Artin and Mumford prove $\operatorname{Tor} H^3(\widetilde{X};\mathbf{Z}) = \mathbf{Z}/2$ for the blown-up double solid $\widetilde{X}$, which is unirational [AM72], [Bea15, §6.3], hence simply connected (Serre). Therefore X is simply connected with $\operatorname{Tor} H^3(X;\mathbf{Z}) = \mathbf{Z}/2$. Universal coefficients and duality give $\operatorname{Tor} H_2 \cong \operatorname{Tor} H^3 = \mathbf{Z}/2$ and $H_3 \cong H^3 = \mathbf{Z}/2$. □

4.3. **Quotient, resolution, fundamental group.** Let $X^\circ \subset X$ be the complement of small G-invariant open balls around the four fixed points, $U = X^\circ/G$, and $\partial U = \bigsqcup_{k=0}^{3} L_k$ with $L_k \cong S^5/(\mathbf{Z}/5)$ lens spaces. Each fixed point has type $\frac{1}{5}(1,1,4)$; let $\rho_k \colon R_k \to \mathbf{C}^3/G$ be the toric resolution given by iterated star subdivision of the cone of $\frac{1}{5}(1,1,4)$ at the lattice points $\frac{1}{5}(1,1,4)$, $\frac{1}{5}(2,2,3)$, $\frac{1}{5}(3,3,2)$, $\frac{1}{5}(4,4,1)$ (Appendix B). It has four exceptional divisors and nine maximal cones; it is a composition of weighted blow-ups, hence projective. Define

$$F = U \cup_{\partial U} \bigsqcup_k R_k = \text{the resolution of } X/G \text{ at its four singular points.}$$

F is a smooth projective threefold. Each R_k is a smooth toric variety whose fan is simplicial with convex full-dimensional support, so its homology is free and concentrated in even degrees [Ful93, §5.2] [**TODO:** page]; it deformation retracts onto the union of the four compact exceptional toric surfaces, with Betti numbers $(1,4,4)$ (Euler characteristic 9 = number of maximal cones).

Lemma 4.4. $\pi_1(F) = 1$.

Proof. X° is simply connected (removing points from a simply connected 6-manifold), so $\pi_1(U) = G$ since G acts freely on X°. A loop in L_k lifting to a path from x to gx in the boundary sphere represents g, so $\pi_1(L_k) \to \pi_1(U)$ is onto. The R_k are simply connected. Van Kampen gives $\pi_1(F) = 1$. (Equivalently, by Armstrong's theorem [Arm68] $\pi_1(X/G) = G/\langle\text{elements with fixed points}\rangle = 1$, and resolving isolated quotient singularities with simply connected exceptional sets does not change π_1.) □

4.4. **Integral homology of F.**

Proposition 4.5. *$H_1(F) = H_5(F) = 0$, $b_1 = b_3 = b_5 = 0$, $b_2(F) = b_4(F) = 19$, $b(F) = 40$, and $\operatorname{Tor} H_i(F) \cong \operatorname{Tor} H_{5-i}(F)$; in particular $\operatorname{Tor} H_4(F) = 0$.*

Proof. By the Lefschetz formula $\chi(X/G) = \frac{1}{5}(\chi(X) + 4 \cdot \#\operatorname{Fix}(g)) = \frac{1}{5}(24+16) = 8$. Each resolution replaces a point by a set of Euler characteristic 9, so $\chi(F) = 8 + 4 \cdot 8 = 40$. Rationally $H_*(U;\mathbf{Q}) = H_*(X^\circ;\mathbf{Q})^G$ and $H_{\mathrm{odd}}(X;\mathbf{Q}) = 0$; with the even homology of the R_k, Mayer–Vietoris gives $b_{\mathrm{odd}}(F) = 0$, hence $b_2 = b_4 = 19$ (the invariant part of $H_2(X;\mathbf{Q})$ has rank 3 by the permutation-module description in Lemma 4.6, and $3+4\cdot 4 = 19$). $H_1(F) = 0$ by Lemma 4.4. F is a closed oriented 6-manifold, so $\operatorname{Tor} H_i \cong \operatorname{Tor} H^{i+1} \cong \operatorname{Tor} H_{5-i}$. □

Lemma 4.6. $\operatorname{Tor} H_2(F) \cong \operatorname{Tor} H_3(F) \cong \mathbf{Z}/2$. *In particular F has no 5-torsion.*

Proof. By Proposition 4.5 it suffices to compute $\operatorname{Tor} H_2(F)$ and $\operatorname{Tor} H_3(F)$ one prime at a time. Mayer–Vietoris for $F = U \cup R$, $R = \bigsqcup R_k$, reads

$$H_i(\partial U) \to H_i(U) \oplus H_i(R) \to H_i(F) \to H_{i-1}(\partial U), \tag{4.1}$$

with $H_*(L_k) = (\mathbf{Z}, \mathbf{Z}/5, 0, \mathbf{Z}/5, 0, \mathbf{Z})$ and $H_*(R)$ free, in even degrees.

Step 1 (input). $\operatorname{Tor} H_2(X) = \mathbf{Z}/2$ and $H_3(X) = \mathbf{Z}/2$ exactly (Proposition 4.3), and $H_i(X^\circ) = H_i(X)$ for $i \le 4$.

Step 2 (localization at $\ell \neq 5$). $H_i(\partial U)_{(\ell)} = 0$ for $i = 1, 2, 3$, so (4.1) gives $H_i(F)_{(\ell)} \cong H_i(U)_{(\ell)} \oplus H_i(R)_{(\ell)}$ for $i = 2, 3$, and $\operatorname{Tor} H_i(F)_{(\ell)} = \operatorname{Tor} H_i(U)_{(\ell)}$.

Step 3 (transfer). As $|G|$ is invertible in $\mathbf{Z}_{(\ell)}$, the Cartan–Leray spectral sequence collapses to $H_i(U; \mathbf{Z}_{(\ell)}) \cong H_i(X^\circ; \mathbf{Z}_{(\ell)})_G \cong H_i(X^\circ; \mathbf{Z}_{(\ell)})^G$ (averaging).

Step 4 (torsion of invariants). For a $\mathbf{Z}_{(\ell)}[G]$-module H with $|G|$ invertible, $H^1(G; \operatorname{Tor} H) = 0$, so $\operatorname{Tor}(H^G) = (\operatorname{Tor} H)^G$. For $\ell = 2$, G acts trivially on $\mathbf{Z}/2$, so $\operatorname{Tor} H_2(F)_{(2)} \cong \operatorname{Tor} H_3(F)_{(2)} \cong \mathbf{Z}/2$. For $\ell \notin \{2, 5\}$ there is no ℓ-torsion.

Step 5 (the prime 5). Let $h \in H^2(X)$ be the pullback of the hyperplane class and $E_1, \dots, E_{10}$ the classes of the exceptional quadrics; g fixes h and permutes the E_i in two free orbits. Pair them with the curve classes $\tilde{\ell}$ (preimage of a general line, $h \cdot \tilde{\ell} = 2$, $E_i \cdot \tilde{\ell} = 0$) and f_i (a ruling of the i-th quadric, $h \cdot f_i = 0$, $E_j \cdot f_i = -\delta_{ij}$, since $E_i|_{E_i} = \mathcal{O}(-1, -1)$). The intersection matrix is $\operatorname{diag}(2, -1, \dots, -1)$, of determinant ± 2. Hence $h, E_1, \dots, E_{10}$ span a sublattice of index dividing 2 in $H^2(X)/\operatorname{Tor}$, and

$$H^2(X; \mathbf{Z}_{(5)}) \cong H_2(X; \mathbf{Z}_{(5)}) \cong \mathbf{Z}_{(5)} \oplus \mathbf{Z}_{(5)}[G]^2$$

as $\mathbf{Z}_{(5)}[G]$-modules (permutation modules are self-dual). In the Cartan–Leray spectral sequence $E^2_{p,q} = H_p(G; H_q(X^\circ; \mathbf{Z}_{(5)})) \Rightarrow H_{p+q}(U; \mathbf{Z}_{(5)})$ [Bro82, p. VII.7] we have $H_1(X^\circ) = 0$, $H_3(X^\circ; \mathbf{Z}_{(5)}) = 0$, and $E^2_{0,2} = (\mathbf{Z} \oplus \mathbf{Z}[G]^2)_G \otimes \mathbf{Z}_{(5)} \cong \mathbf{Z}^3_{(5)}$. Since $H_2(G; \mathbf{Z}) = 0$, the only possible differential into $E_{0,2}$ comes from the finite group $E^3_{3,0} = H_3(G; \mathbf{Z}) = \mathbf{Z}/5$; it is zero because $E^2_{0,2} \cong \mathbf{Z}^3_{(5)}$ is torsion free. Hence $H_2(U; \mathbf{Z}_{(5)}) \cong \mathbf{Z}^3_{(5)}$. Since moreover $b_3(U) = 0$, universal coefficients give $H^3(U; \mathbf{Z}) \otimes \mathbf{Z}_{(5)} = 0$, and Lefschetz duality (U is a compact oriented 6-manifold with boundary) gives $H_3(U, \partial U)_{(5)} \cong H^3(U)_{(5)} = 0$. Hence $H_3(\partial U) \to H_3(U)$ is onto after localizing at 5. As $H_3(R) = 0$ and $H_2(\partial U) = 0$, (4.1) gives $H_3(F)_{(5)} = \operatorname{coker}(H_3(\partial U) \to H_3(U))_{(5)} = 0$.

Step 6 (duality at 5). From (4.1) alone, $H_2(F)_{(5)}$ is an extension of $\ker\big(H_1(\partial U) \to H_1(U)\big) \cong (\mathbf{Z}/5)^3$ by the free module $H_2(U)_{(5)} \oplus H_2(R)_{(5)}$, so Mayer–Vietoris does not by itself exclude a $\mathbf{Z}/5$ in $H_2(F)$. Torsion-freeness comes from duality: $\operatorname{Tor} H_2(F)_{(5)} \cong \operatorname{Tor} H_3(F)_{(5)} = 0$ by Step 5. □

Remark 4.7. The exactness of Lemma 4.6 is essential. If $\operatorname{Tor} H_2(F) = (\mathbf{Z}/2)^2$, or if F had 5-torsion in degrees 2, 3, the gap of Theorem A (and of Remark 7.1) would collapse to ≤ 0 (`computations/homology/w10.py`, `computations/homology/lam.py`). Note that $H_3(U; \mathbf{Z}_{(5)})$ itself has order 25 (contributions $E^\infty_{1,2}$ and $E^\infty_{3,0}$); it is killed in F by the lens space boundary.

4.5. **A Godeaux-type divisor.** The G-invariant quintic monomials (weight sum $\equiv 0 \bmod 5$) span a 12-dimensional space. Fix the invariant quintic

$$\begin{aligned} Q = {} & 6x_0^5 + 9x_1^5 + 17x_0x_1^3x_2 + 15x_0^2x_1x_2^2 + 13x_2^5 + 14x_0^3x_1x_3 + 10x_1^2x_2^2x_3 \\ & + 10x_0x_2^3x_3 + x_0x_1^2x_3^2 + 19x_0^2x_2x_3^2 + 15x_1x_2x_3^3 + 5x_3^5. \end{aligned}$$

Lemma 4.8 (machine-verified, Appendix B)**.** *The surface $\{Q = 0\}$ is smooth, avoids $p_0, \dots, p_3$ and the ten nodes of B, and meets B transversally in a smooth curve Δ.*

Let $\widetilde{S} \subset X$ be the preimage of $\{Q = 0\}$; it is the double cover of the quintic surface branched along $\Delta \in |4H|$, and lies in the locus where $X \to \overline{X}$ is an isomorphism and G acts freely. Put $S = \widetilde{S}/G \subset F$.

Proposition 4.9. *S is a smooth divisor in F with $\pi_1(S) = \mathbf{Z}/5$; its homology is*

$$H_*(S; \mathbf{Z}) = (\mathbf{Z},\ \mathbf{Z}/5,\ \mathbf{Z}^{40} \oplus \mathbf{Z}/5,\ 0,\ \mathbf{Z}).$$

Proof. Smoothness and the divisor property follow from Lemma 4.8 and the fact that $\widetilde{S}$ avoids the fixed points. Simple connectivity of $\widetilde{S}$: by Nori's theorem [Nor83], since Δ is a smooth curve with $\Delta^2 > 0$ on the simply connected quintic surface, π_1 of the complement of Δ is abelian, hence cyclic generated by a meridian γ; the double cover corresponds to $\langle\gamma^2\rangle$, which is generated by a meridian of $\widetilde{\Delta}$ and dies in $\widetilde{S}$. Here $H_1(\Sigma \setminus \Delta) = \mathbf{Z}/4$, because the hyperplane class H of the quintic surface Σ is primitive ($H^2 = 5$ is squarefree), so $\Delta \in |4H|$ has divisibility exactly 4. **[TODO:** theorem number in [Nor83].] Hence $\pi_1(S) = G = \mathbf{Z}/5$, $H_1(S) = \mathbf{Z}/5$, $\operatorname{Tor} H_2(S) \cong \operatorname{Tor} H_1(S) = \mathbf{Z}/5$ and $H_3(S) = 0$. For Betti numbers, χ of a smooth quintic surface is 55 and Δ has $2g - 2 = \Delta \cdot (\Delta + K) = 4H \cdot 5H = 100$, so $\chi(\widetilde{S}) = 2 \cdot 55 + 100 = 210$, $\chi(S) = 42$ and $b_2(S) = 40$. □

We call S *Godeaux-type* because $\pi_1(S) = \mathbf{Z}/5$; it is not a Godeaux surface (its geometric genus is positive).

Remark 4.10. This gives, in the Kähler category, a simply connected compact Kähler threefold with 2-primary Brauer-type torsion containing a smooth divisor with 5-torsion in H_1. The Brauer group is a birational invariant, which is why one cannot start from a rational threefold such as the toric resolution of $\mathbf{P}^3/\mathbf{Z}_5$.

5. The ten-dimensional construction

5.1. **The surface Y and two invariant lines.** Let Y be $\mathbf{CP}^1 \times \mathbf{CP}^1$ blown up at the four points $\{0,\infty\}^2$, with the diagonal circle action $t \cdot (z,w) = (e^{2\pi i t}z, e^{2\pi i t}w)$ lifted to Y and an invariant Kähler form. By [OpenAI22, Prop. 4.3], the half-turn g_Y is a Hamiltonian involution with $\operatorname{Fix}(g_Y) = E_{00} \sqcup E_{0\infty} \sqcup E_{\infty 0} \sqcup E_{\infty\infty}$ (the exceptional curves), $dg_Y = +1$ on TE and -1 on the normal line; Y is simply connected with $H_*(Y) = (\mathbf{Z}, 0, \mathbf{Z}^6, 0, \mathbf{Z})$; and $c_1(TY)[E] = 1$.

Let $L_0, L_2 \subset Y$ be the proper transforms of $\{0\} \times \mathbf{CP}^1$ and $\{\infty\} \times \mathbf{CP}^1$. They are disjoint invariant smooth rational curves; g_Y rotates each by π, with fixed points ("poles") $L_0 \cap E_{00}$, $L_0 \cap E_{0\infty}$, $L_2 \cap E_{\infty 0}$, $L_2 \cap E_{\infty\infty}$. Thus every fixed curve contains exactly one pole, and at a pole y the curves L and E meet transversally, so $T_yY = T_yE \oplus T_yL$ with $dg_Y = \operatorname{diag}(1,-1)$.

5.2. **Transverse center lemma.**

Lemma 5.1 (transverse center lemma)**.** *Let A be a complex manifold, g a holomorphic automorphism of finite order, $\operatorname{Fix}(g) = \bigsqcup_j C_j$, and $W \subset A$ a g-invariant closed complex submanifold of codimension ≥ 2. Assume that for every $p \in W \cap \operatorname{Fix}(g)$*

$$E_p^{\neq} := \operatorname{im}(d_pg - I) \subset T_pW. \tag{5.1}$$

Let $\hat{g}$ be the lift of g to $\widehat{A} = \operatorname{Bl}_W A$. Then

$$\operatorname{Fix}(\hat{g}) = \bigsqcup_j \widehat{C}_j, \qquad \widehat{C}_j \cong \operatorname{Bl}_{W \cap C_j} C_j,$$

where $\widehat{C}_j$ is the proper transform of C_j (equal to C_j if $W \cap C_j = \varnothing$, and to C_j blown up along a smooth center otherwise). In particular $\hat{g}$ has no fixed components other than the proper transforms of those of g.

Proof. Write $\pi\colon \widehat{A} \to A$ and $E = \pi^{-1}(W) = \mathbf{P}(N_W)$.

(a) Cleanness. $W \cap \operatorname{Fix}(g) = \operatorname{Fix}(g|_W)$ is a closed submanifold of W with $T_p(W \cap C_j) = \ker(d_pg - I) \cap T_pW = T_pC_j \cap T_pW$; so W and each C_j intersect cleanly. By (5.1), $T_pA = T_pC_j \oplus E_p^{\neq} \subset T_pC_j + T_pW$, so $T_pA = T_pC_j + T_pW$.

(b) Fixed points off E, and over non-fixed points of W. π is a g-equivariant isomorphism off E, so $\operatorname{Fix}(\hat{g}) \setminus E = \operatorname{Fix}(g) \setminus W$. If $p \in W$ and $g(p) \neq p$, then $\hat{g}$ maps the fiber E_p to $E_{g(p)} \neq E_p$, so it has no fixed points there.

(c) Fibers over $W \cap \mathrm{Fix}(g)$. For $p \in W \cap C_j$, $\hat{g}$ acts on $E_p = \mathbf{P}(T_pA/T_pW)$ through the map induced by d_pg. By (5.1), T_pA/T_pW is the image of $E_p^+ = T_pC_j$, on which $d_pg = \mathrm{id}$. Hence $\hat{g}$ fixes E_p pointwise.

(d) Identification. By (a) the natural map $T_pC_j/T_p(W \cap C_j) \to T_pA/T_pW$ is injective (cleanness) and surjective ($T_pA = T_pC_j + T_pW$), so $\mathbf{P}(N_{W\cap C_j/C_j})|_p \to E_p$ is an isomorphism. By cleanness the scheme-theoretic intersection $W \cap C_j$ is the smooth submanifold above, so the strict transform $\widehat{C}_j$ (the closure of $C_j \setminus W$) is isomorphic to $\mathrm{Bl}_{W\cap C_j} C_j$ [Har77, p. II.7.15]; it is a closed connected submanifold whose intersection with E is its exceptional divisor $\mathbf{P}(N_{W\cap C_j/C_j})$, which by the isomorphism just noted is all of $E|_{W\cap C_j}$. It is contained in the closed set $\mathrm{Fix}(\hat{g})$. Combining (b) and (c), as sets

$$\mathrm{Fix}(\hat{g}) = (\mathrm{Fix}(g) \setminus W) \cup E|_{W\cap \mathrm{Fix}(g)} = \bigsqcup_j ((C_j \setminus W) \cup E|_{W\cap C_j}) = \bigsqcup_j \widehat{C}_j.$$

The $\widehat{C}_j$ are disjoint because the C_j are, and $\mathrm{Fix}(\hat{g})$, a closed complex submanifold, is their disjoint union. □

Remark 5.2. Without (5.1) the lemma fails: blowing up $S \times \{y\}$ with y a pole, or $S \times E$, produces additional fixed components (copies of S or $S \times \mathbf{CP}^1$ inside E) that carry torsion and destroy the gap; this is a direct computation with the bookkeeping of `computations/homology/lam.py`. See also §6.4.

5.3. **The manifold M and its fixed locus.** Let F and $S \subset F$ be as in §4 and put

$$W = S \times (L_0 \sqcup L_2) \subset F \times Y, \qquad M = \mathrm{Bl}_W(F \times Y).$$

W has complex dimension 3, hence codimension 2 in the fivefold $F \times Y$.

Proposition 5.3. *M is a simply connected compact Kähler manifold of complex dimension 5 with a Hamiltonian circle action lifting $\mathrm{id}_F \times$ (diagonal action). Its half-turn $\hat{g}$ is a Hamiltonian involution with*

$$\mathrm{Fix}(\hat{g}) = C_1' \sqcup C_2' \sqcup C_3' \sqcup C_4', \qquad C_j' \cong \mathrm{Bl}_{S\times\{\mathrm{pt}\}}(F \times \mathbf{CP}^1),$$

each C_j' simply connected of real dimension 8. The minimal Chern number of M is 1.

Proof. Kählerness and simple connectivity follow from Proposition 2.7. W is invariant, so the holomorphic circle action lifts; averaging a Kähler form and using $H^1(M;\mathbf{R}) = 0$ makes it Hamiltonian, and its half-turn $\hat{g}$ is a Hamiltonian involution lifting $g = \mathrm{id}_F \times g_Y$. We have $\mathrm{Fix}(g) = \bigsqcup_j F \times E_j$ and $W \cap \mathrm{Fix}(g) = S \times \{\text{poles}\}$. At $p = (s, y)$ with y a pole, $E_p^{\neq} = \{0\} \oplus T_yL \subset T_sS \oplus T_yL = T_pW$, so (5.1) holds. Lemma 5.1 gives $\mathrm{Fix}(\hat{g}) = \bigsqcup \widehat{F \times E_j}$ with $\widehat{F \times E_j} \cong \mathrm{Bl}_{S\times\{y_j\}}(F \times E_j)$, y_j the unique pole on E_j. These are simply connected by Proposition 2.7. Finally, for $x \in F \setminus S$ the sphere $\{x\} \times E_{00}$ is disjoint from W and has $c_1 = 1$. □

5.4. **Integral homology and torsion profiles.** By Proposition 2.7 (codimension 2 in both cases),

$$H_*(M) \cong H_*(F \times Y) \oplus 2\,H_*(S \times \mathbf{CP}^1)[2], \qquad H_*(C_j') \cong H_*(F \times \mathbf{CP}^1) \oplus H_*(S)[2]. \tag{5.2}$$

All Künneth factors other than F, S have free homology, so no Tor terms arise. Write $a = b_2(F)$ and $\beta = b_2(S)$.

Derivation of the tables. F has $\mathrm{Tor}\,H_2 = \mathrm{Tor}\,H_3 = \mathbf{Z}/2$; Künneth with Y (free ranks $1, 6, 1$ in degrees $0, 2, 4$) places 2-torsion of multiplicities $(1, 6, 1)$ in degrees $(2, 4, 6)$ and $(3, 5, 7)$. $S \times \mathbf{CP}^1$ has 5-torsion $\mathbf{Z}/5$ in degrees $1, 2, 3, 4$; shifted by 2 and doubled, this gives $(2, 2, 2, 2)$ in degrees $3, \dots, 6$. For C_j': $F \times \mathbf{CP}^1$ has 2-torsion in degrees $2, 3, 4, 5$, and $H(S)[2]$ adds $\mathbf{Z}/5$ in degrees $3, 4$. The free ranks follow in the same way from $r(F) = (1, a, a, 1)$, $r(S) = (1, \beta, 1)$. □

	M						C_j'			
degree	2	3	4	5	6	7	2	3	4	5
d_2	1	1	6	6	1	1	1	1	1	1
d_5	0	2	2	2	2	0	0	1	1	0
$t_i = \max$	1	2	6	6	2	1	1	1	1	1

TABLE 2. Primary torsion multiplicities $d_p(\mathrm{Tor}\, H_i)$. All torsion is elementary abelian. $\sum_i t_i(M) = 18$, $\sum_i t_i(C_j') = 4$.

degree	0	2	4	6	8	10	total
$r_i(M)$	1	$a+8$	$7a+2\beta+3$	$7a+2\beta+3$	$a+8$	1	$16a+4\beta+24$
$r_i(C_j')$	1	$a+2$	$2a+\beta$	$a+2$	1		$4a+\beta+6$

TABLE 3. Free ranks (all odd ranks vanish). $b(M) = 4b(C_j')$.

Corollary 5.4.

$$\lambda(M) = b(M) + 36, \qquad \lambda(C_j') = \tfrac{1}{4}b(M) + 8, \qquad \lambda(M) - \sum_{j=1}^{4} \lambda(C_j') = 4,$$

independently of a and β. Moreover $\lambda_1(M) = b(M) + 32 = \sum_j \lambda(C_j')$.

Proof. The first three identities follow from the tables. For λ_1, the even-degree torsion of M has 2-multiplicity $1+6+1 = 8$ and 5-multiplicity $2+2 = 4$, so $d = 8$; likewise in odd degrees. Thus $\lambda_1(M) = b(M) + 2(8+8)$. □

The source of the gap is visible in Table 2: in M the dominant prime is 5 in degrees $3, 6$ and 2 in degrees $4, 5$, so both primes contribute; in C_j' the 2- and 5-torsion occupy the same degrees $3, 4$ with equal multiplicity and are absorbed by the maximum. The script `computations/homology/m10_indep.py` reproduces these numbers. It hard-codes placeholder free ranks $b(F) = 8$, $b(S) = 11$ and prints $\lambda(M) = 144$, $\sum \lambda(C_j') = 140$, cyclic bound 140; by Corollary 5.4 the gap 4 is independent of these placeholders (also tested for $b(F) \in \{8, 22\}$, $b_2(S) \in \{9, 30\}$). The actual values are given in Remark 5.5.

Remark 5.5. With $a = 19$ (Proposition 4.5) and $\beta = 40$ (Proposition 4.9): $b(M) = 488$, $\lambda(M) = 524$, $\lambda(C_j') = 130$, $\#\,\mathrm{Fix} = 520$, and the relative deficit is $4/524 = 1/131$. The value $b(F) = 40$ was obtained twice, from two independent constructions of the toric fan (seven rays, nine smooth maximal cones; `computations/sage/bF2.sage`); it depends on the choice of resolution, the gap does not. Over $\mathbf{F}_2$ one has $b(M; \mathbf{F}_2) = b(M) + 2 \cdot 16 = \#\,\mathrm{Fix}$, so the count also equals the total mod-2 Betti number; over $\mathbf{F}_5$, $b(M; \mathbf{F}_5) = b(M) + 16 < \#\,\mathrm{Fix}$.

5.5. **Proof of Theorem A.** By Theorem 2.3 with $D = 8$, each C_j' carries a Morse function h_j with exactly $\lambda(C_j')$ critical points. Proposition 2.6 applied to the Hamiltonian involution $\hat{g}$ of Proposition 5.3 yields a one-periodic Hamiltonian H on M with $\#\,\mathrm{Fix}(\phi_H^1) = \sum_j \lambda(C_j')$, all nondegenerate, all with contractible orbits since M is simply connected. By Corollary 5.4 this equals $\lambda(M) - 4$, and $\lambda(M) = \mathrm{SM}(M) = \mathrm{Morse}(M)$ by Theorem 2.3 with $D = 10 \geq 9$ ([OpenAI22, Thm. 2.4]). The minimal Chern number is 1 (Proposition 5.3) and $\#\,\mathrm{Fix} = \lambda_1(M)$ by Corollary 5.4. □

Remark 5.6. Theorem A is consistent with Theorem 2.4, which here is an equality. It shows that the cyclic integral Floer bound is sharp in complex dimension 5 (previously complex dimension 1666 [OpenAICompanion]).

6. Dimension eight: necessary conditions and a mechanism obstruction

6.1. Necessary torsion conditions.

Proposition 6.1. *Let M be a simply connected closed symplectic 8-manifold with minimal Chern number N, and put $T_i = \operatorname{Tor} H_i(M)$. Then $T_i \cong T_{7-i}$, $T_i = 0$ for $i \notin \{2,3,4,5\}$, $\lambda_N(M) = \lambda(M)$ for $N \geq 2$ (including $N = \infty$), and*

$$\lambda(M) - \lambda_1(M) = 4\big[d(T_2) + d(T_3) - d(T_2 \oplus T_3)\big].$$

Consequently, assuming Theorem 2.4 and Theorem 2.3 for $D = 8$, a nondegenerate Hamiltonian on M with fewer than $\mathrm{SM}(M)$ fixed points can exist only if $N = 1$ and T_2, T_3 are nonzero with different dominant primes in the sense of (2.1). In particular $\operatorname{Tor} H^3(M) \cong T_2 \neq 0$.

Proof. $T_i \cong \operatorname{Tor} H^{i+1} \cong T_{7-i}$ by universal coefficients and duality; $T_0 = T_1 = 0$ forces $T_6 = T_7 = 0$. If $2N \geq 4$ the degrees $2,3,4,5$ lie in distinct classes mod $2N$. For $N = 1$ the even class carries $T_2 \oplus T_4 \cong T_2 \oplus T_3$ and the odd class $T_3 \oplus T_5 \cong T_3 \oplus T_2$; compare with λ, which counts $d(T_2) + d(T_3)$ twice in each parity. The rest follows from Theorems 2.4 and 2.3 and (2.1). □

Remark 6.2. Blowing up a space with free homology along any centers never creates torsion in degree 2 (the shifts in Proposition 2.7 are ≥ 2 and H_0 of a center is free). Heuristically, therefore, a blow-up construction in dimension 8 needs an ambient with Brauer-type torsion from the start; this remark applies only to constructions by blow-ups.

Proposition 6.3 (failure of the product mechanism below dimension 12)**.** *Let X be a simply connected closed Kähler manifold of real dimension ≤ 6 and Y as in §5. Then $\lambda(X \times Y) = 4\lambda(X \times \mathbf{CP}^1)$.*

Proof. If $\dim_{\mathbf{R}} X \leq 4$, X has no torsion. If $\dim_{\mathbf{R}} X = 6$, the proof of Proposition 3.1 shows that the only torsion is $T := T_2 \cong T_3$. Then $X \times \mathbf{CP}^1$ has torsion T in degrees $2,3,4,5$ and $X \times Y$ has T, T, T^6, T^6, T, T in degrees $2, \ldots, 7$, so $\sum t_i(X \times Y) = 16d(T) = 4\sum t_i(X \times \mathbf{CP}^1)$; free ranks scale by 4 as well. □

6.2. Perfectness of moment maps over prime fields.

Lemma 6.4. *Let C be a closed symplectic manifold with a Hamiltonian circle action, moment map μ, and fixed components F of Morse–Bott index $2\mu_F$. For every field k and every i,*

$$\dim H_i(C;k) = \sum_F \dim H_{i-2\mu_F}(F;k).$$

Proof. Let q be a prime. "$\leq$": the Morse–Bott inequalities for μ hold with $\mathbf{F}_q$ coefficients, since negative normal bundles are complex, hence oriented, so the Thom isomorphism applies. "$\geq$" for total dimensions: let $X_n = C^{\mathbf{Z}/q^n}$, where $\mathbf{Z}/q^n \subset S^1$. The group $\mathbf{Z}/q^{n+1}$ preserves X_n and acts on it through $\mathbf{Z}/q^{n+1}/\mathbf{Z}/q^n \cong \mathbf{Z}/q$ with fixed set X_{n+1}. Smith theory for $\mathbf{Z}/q$ [Bre72, Ch. III] gives $\dim H_*(X_{n+1};\mathbf{F}_q) \leq \dim H_*(X_n;\mathbf{F}_q)$. A smooth action of S^1 on a compact manifold has finitely many isotropy groups; since $\bigcup_n \mathbf{Z}/q^n$ is dense in S^1, $X_n = C^{S^1}$ for n large. Hence $\sum_F \dim H_*(F;\mathbf{F}_q) \leq \dim H_*(C;\mathbf{F}_q)$. Equality of total dimensions forces the Morse–Bott spectral sequence (with E^1 the shifted homology of the F's) to degenerate, giving degreewise equality. For k of characteristic 0, apply the result for a prime q that divides no torsion order of C or of any F (or use [Fra59; Kir84]). The case of a general field reduces to its prime field. □

This is the classical Borel–Smith inequality for torus actions over $\mathbf{F}_p$ [AP93, Ch. 3], [Hsi75, Ch. III], specialized to moment maps; we do not claim novelty for Lemma 6.4. [**TODO:** precise theorem numbers in Allday–Puppe and Hsiang; literature search for prior explicit statements of $\mathbf{F}_p$-perfectness of moment maps.]

Corollary 6.5. *In the situation of Lemma 6.4, for every prime p and every i,*

$$r_i(C) = \sum_F r_{i-2\mu_F}(F), \qquad t_i^p(C) = \sum_F t_{i-2\mu_F}^p(F).$$

Proof. $\dim H_i(\,\cdot\,;\mathbf{F}_p) = r_i + t_i^p + t_{i-1}^p$. Subtract the rational identity and induct on i starting from $t_{-1}^p = 0$. □

6.3. **Proof of Theorem B.** Choose a circle $S^1 \subset T$ with $M^{S^1} = M^T$: T has finitely many isotropy subgroups on M, and a circle whose Lie algebra avoids the finitely many proper subspaces $\mathrm{Lie}(H)$, $H \neq T$ an isotropy group, works. Let μ be its moment map. Each component C of $\mathrm{Fix}(g)$ is a T-invariant symplectic submanifold, $\mu|_C$ is a moment map for the circle on C, and $C^{S^1} = C \cap M^T$. Write $\mathcal{F}$ for the set of components of M^T and $\mathcal{F}(C) \subset \mathcal{F}$ for those in C; $\mathcal{F} = \bigsqcup_C \mathcal{F}(C)$. For $F \in \mathcal{F}(C)$ write $2\nu_F$ and $2\mu_F$ for its indices in C and in M.

Step 1: an upper bound for $\lambda(M)$. By Corollary 6.5 applied to M, $b(M) = \sum_F b(F)$ and

$$t_i(M) = \max_p \sum_F t_{i-2\mu_F}^p(F) \le \sum_F \max_p t_{i-2\mu_F}^p(F) = \sum_F t_{i-2\mu_F}(F).$$

Summing over i: $\lambda(M) \le \sum_{F\in\mathcal{F}} \lambda(F)$.

Step 2: equality for each C. We claim $\lambda(C) = \sum_{F\in\mathcal{F}(C)} \lambda(F)$ for every component C of $\mathrm{Fix}(g)$; by hypothesis $\dim_{\mathbf{R}} C \le 6$. (If $\dim_{\mathbf{R}} M = 8$, this holds unless g acts trivially on M, in which case $C = M$ and the theorem is trivial.) Nothing below uses $\dim M$ or $\pi_1(C)$. A component F with nonzero torsion has $\dim_{\mathbf{R}} F \ge 4$ (closed orientable manifolds of dimension ≤ 2 have free homology). If $F = C$ there is nothing to prove. Otherwise $\dim_{\mathbf{R}} C = 6$ and $\dim_{\mathbf{R}} F = 4$, and the normal bundle of F in C is a complex line, so $\nu_F \in \{0, 1\}$ and F is a local minimum or local maximum of $\mu|_C$. By the connectedness theorem of Atiyah and Guillemin–Sternberg [Ati82; GS82], local extrema are global and the minimum and maximum components are unique. Hence C contains at most two torsion-carrying fixed components: the minimum F_- ($\nu = 0$, torsion in degrees $1, 2$) and the maximum F_+ ($\nu = 1$, torsion in degrees $3, 4$). These degree supports are disjoint, so by Corollary 6.5 each $t_i(C)$ equals $t_i(F_-)$ or $t_{i-2}(F_+)$ or 0, whence $\sum_i t_i(C) = \sum_{F\in\mathcal{F}(C)} \sum_i t_i(F)$. Together with $b(C) = \sum_{F\in\mathcal{F}(C)} b(F)$ this proves the claim.

Step 3. $\sum_j \lambda(C_j) = \sum_j \sum_{F\in\mathcal{F}(C_j)} \lambda(F) = \sum_{F\in\mathcal{F}} \lambda(F) \ge \lambda(M)$.

Consequence. The fixed points of $\psi^\epsilon \circ g$ are the critical points of Morse functions h_j on C_j, and $\#\,\mathrm{Crit}(h_j) \ge \lambda(C_j)$ by Proposition 2.2. Hence $\#\,\mathrm{Fix} \ge \lambda(M)$, which equals $\mathrm{SM}(M)$ for simply connected M by Theorem 2.3 ($D = 8$). □

Remark 6.6 (a support criterion). The proof shows, in any dimension, that $\sum_j \lambda(C_j) \ge \lambda(M)$ whenever, in every C_j, the index-shifted torsion supports of the torsion-carrying fixed components are pairwise disjoint. A construction of this type in dimension $2n$ therefore needs a component C_j with two torsion-carrying T-fixed components whose shifted supports overlap. In Theorem A this happens as follows. The diagonal circle has weights $(1, 1)$ at $(0, 0)$ and $(-1, -1)$ at (∞, ∞), so it acts *trivially* on E_{00} and $E_{\infty\infty}$; hence C_1' and C_4' (over these curves) are themselves T-fixed components. On C_2' and C_3' (over $E_{0\infty}$, $E_{\infty 0}$) the circle rotates the $\mathbf{CP}^1$-factor, and the T-fixed components are three: the proper transform of $F \times \{y_j\}$ ($\cong F$, since $S \subset F$ is a divisor), $F \times \{y_j'\}$ for the other fixed point y_j' of E_j, and the section $\mathbf{P}(T_{y_j} E_j) \cong S$ of the exceptional divisor (the section $\mathbf{P}(N_{S/F})$ lies in the proper transform of $F \times \{y_j\}$). As a check, the total Betti number of M^T is $2 \cdot 122 + 2\,(40+40+42) = 488 = b(M)$, as Lemma 6.4 requires. Within C_2' one has $\sum_F \lambda(F) = 44+44+46 = 134 > 130 = \lambda(C_2')$: the loss of 4 comes from the overlap of the 5-torsion of S (degrees $3, 4$ after the index shift) with the 2-torsion of the copies of F (degrees 2–5). Altogether $\sum_{F\subset M^T} \lambda(F) = 528 \ge \lambda(M) = 524 \ge \sum_j \lambda(C_j') = 520$, consistent with Step 1.

Remark 6.7 (a second argument via Li's theorem). By a theorem of H. Li [Li03, Thm. 0.1], for a Hamiltonian circle action on a compact symplectic manifold C, $\pi_1(C) \cong \pi_1(F_{\min}) \cong \pi_1(F_{\max}) \cong \pi_1(C_{\mathrm{red}})$. If $\dim C = 6$ and C is simply connected, its four-dimensional fixed components (necessarily extremal) are simply connected, hence torsion-free, and the torsion-carrying fixed components of M are themselves components of $\mathrm{Fix}(g)$; the inequality of Step 1 then gives the claim. If some C_j is not simply connected, Lemma 6.4 prevents cancellation of its H_2-torsion. This gives an alternative route to Theorem B; we have not written it out in full.

6.4. **Dimensional limitation of transverse centers (heuristic).** The following is a heuristic, not a theorem. In Theorem A the extra torsion of the fixed components is introduced by a blow-up whose center W meets $\mathrm{Fix}(g)$ cleanly and satisfies (5.1). If the inserted torsion is that of a surface (S, real dimension 4, with odd-prime torsion in H_1), then W must contain S and, by (5.1), the normal directions $E^{\neq}$ of the fixed component (real dimension ≥ 2), and must have real codimension ≥ 4 (a divisor blow-up is the identity). Hence $\dim_{\mathbf{R}} M \geq 4 + 2 + 4 = 10$. The argument assumes that the inserted torsion comes from a surface contained in the center; it is independent of, and weaker than, Theorem B.

7. Higher dimensions, earlier claims, and open problems

7.1. **Kähler examples in dimensions twelve and sixteen.**

Remark 7.1 (dimension 12). Let $X^8 = \mathrm{Bl}_{S\times\{t_1\}\sqcup S\times\{t_2\}}(F \times \mathbf{CP}^1)$ and $M^{12} = X^8 \times Y$, $C = X^8 \times \mathbf{CP}^1$. Then M^{12} is simply connected Kähler, and the product mechanism gives $\lambda(M^{12}) - 4\lambda(C) = 8$; with $b(F) = 40$, $b(S) = 42$ one gets $\lambda(M^{12}) = 1400$ and $\#\,\mathrm{Fix} = 1392$. The gap relies on Lemma 4.6 (no 5-torsion in F); with 5-torsion of rank ≥ 1 in F it would vanish.

Remark 7.2 (dimension 16). In $P = \mathbf{CP}^1 \times \mathbf{CP}^5$ embed disjointly (by [OpenAI22, Lemma 3.2]) a bielliptic surface S_2 with $(\mathbf{Z}/2)^2$ torsion and three bielliptic surfaces $S_3^{(i)}$ with $\mathbf{Z}/3$ torsion into $\mathbf{CP}^5$, and blow up P along $S_2 \times \mathbf{CP}^1$ (codimension 3) and $S_3^{(i)} \times \{t\}$ (codimension 4). For the resulting complex sixfold X and $M = X \times Y$ one finds $\lambda(M) = 1232$, $4\lambda(X \times \mathbf{CP}^1) = 1216$ (two independent scripts, `computations/homology/indep.py` and `computations/homology/x12.py`); the family with $3m$ copies of S_3 and m copies of $S_2 \times \mathbf{CP}^1$ has $\lambda = 96 + 1136m$ and gap $16m$. Here $\#\,\mathrm{Fix} = 1216$ also equals the Bai–Xu cyclic bound $\lambda_1(M)$ (minimal Chern number 1), as recomputed with `computations/homology/indep.py`. This is a rearrangement of the centers of [OpenAI22] and is not claimed as a separate result.

7.2. **Relation to an earlier claim.** Ma's preprint [Ma13, Thm. 1.5] asserts that the minimal number of fixed points of a nondegenerate Hamiltonian diffeomorphism of a closed symplectic manifold equals the ordinary Morse number (see also [Vib], cited for context only). This assertion is already refuted by [OpenAISept23, Thm. 1.1], on a closed symplectic manifold of real dimension 12 with $\pi_1 = A_5^k$, which is not simply connected, and by the simply connected examples of [OpenAIStable; OpenAI22; OpenAICompanion] in higher dimensions. For the manifold M of Theorem A, $\mathrm{Morse}(M) = \mathrm{SM}(M) = \lambda(M)$ by Theorem 2.3, while $\#\,\mathrm{Fix}(\phi_H^1) = \lambda(M) - 4$. Theorem A is thus the first simply connected example below the stable Morse number in real dimension less than 22, and it refutes the assertion of [Ma13] on a simply connected Kähler manifold of real dimension 10. [**TODO:** we have not read [Ma13] directly; the theorem number and statement are taken from [OpenAISept23].]

7.3. **Open problem: dimension eight.** By Theorems A and B and Proposition 3.1, among simply connected closed symplectic manifolds of real dimension at least 6 the minimal real dimension of a counterexample to (1.1) is 8 or 10 (Corollary 1.1). Simply connected real dimension 4 remains open in the sense of §3, and non-simply-connected manifolds of real dimension 4, 6, 8, 10 are not addressed here.

Problem 7.3. Does there exist a closed (simply connected, Kähler) symplectic 8-manifold M and a nondegenerate Hamiltonian diffeomorphism with fewer than $\mathrm{SM}(M)$ fixed points?

By Proposition 6.1, such an M must have $N = 1$ and Brauer-type torsion $T_2 \neq 0$ whose dominant primes differ from those of T_3. By Theorem B, the Hamiltonian cannot be obtained by perturbing a finite-order element of a Hamiltonian torus action. A positive answer requires either a finite-order Hamiltonian diffeomorphism not contained in any Hamiltonian torus action, or an entirely different mechanism. A negative answer requires a Floer-theoretic obstruction stronger than Theorem 2.4, which by Proposition 6.1 is not sufficient in general.

Appendix A. Computational corroboration

The script `computations/milp/lpsearch2.py` formulates a mixed-integer linear program modelling the torsion bookkeeping of the finite-order/torus mechanism in real dimension $2n$. Variables are nonnegative weights x of "torsion pieces" (prime $\in \{p, q\}, f, (a, b), c, j, \nu, \varepsilon$): a pair of Poincaré-dual cyclic summands in degrees $a < b$ of a T-fixed component of complex dimension f, contained in a component C of $\mathrm{Fix}(g)$ of complex dimension c with label $j \leq K$, with index 2ν in C and $2(\nu + \varepsilon)$ in M. Constraints: Poincaré duality of the torsion profiles of M and of each C; no torsion of M in degrees $0, 1$; uniqueness of minimum and maximum components in each C and globally; if $F = C$ then C contains no other piece; total weight ≤ 1. The objective is the normalized gain $\sum_k \max(\mathrm{profile}_M(k)) - \sum_C \sum_k \max(\mathrm{profile}_C(k))$, linearized with binary selectors. The modelling of the torsion of C as the sum of the shifted torsion of its fixed components is justified by Corollary 6.5, so no extra perfectness hypothesis is needed.

real dim	setting	max normalized gain
8	$K = 2, 3$; with or without H_1-torsion in C	0
10	$K = 2, 3$, no H_1-torsion in C	1/3
10	$K = 3$ unrestricted	1/2
12	$K = 2, 3$	1 (positive control)
10	forbid 6-dim F / Brauer-type $(2, 3)$ F / surfaces / 8-dim C	0 in each case

Table 4. Results (`computations/milp/run_lp.py`, `computations/milp/necessity.py`).

In dimension 10 the optimal configurations contain an 8-dimensional C whose minimum and maximum (or saddles) are 6-dimensional Brauer-type components and which contains torsion surfaces of the other prime at index 2; by Guillemin–Sternberg birational cobordism [GS89], such a surface is a codimension-2 symplectic submanifold of a reduced space of Brauer type, which led to the construction of §4. *Caveats:* this is an LP relaxation with real weights and two primes, it encodes only the listed necessary conditions and not realizability, and it is used only as corroboration; Theorem B has a complete hand proof.

Appendix B. Machine verification of the symmetroid and the quintic

All paths are relative to the repository root. In exact arithmetic over $\mathbf{Q}$ (and over $\overline{\mathbf{Q}}$ where stated):

- `computations/sage/q10.sage` (random seed 7) produces the web and the quintic and verifies: the character $f(gx) = \zeta^3 f(x)$; irreducibility of f over $\mathbf{Q}$; that the saturated Jacobian ideal of f has Hilbert polynomial 10 and equals its radical and the saturated ideal of 3×3 minors of A; that the coordinate points lie on B and are smooth points (gradients $(0, 0, -1728, 0)$, $(0, 0, 0, -875)$, $(0, -640, 0, 0)$, $(-672, 0, 0, 0)$ at $p_0, \ldots, p_3$); that the web has no base points; that a general (non-invariant) web also has total Tjurina number 10; that the invariant quintic Q is smooth, avoids the coordinate points and the nodes, and that $B \cap \{Q = 0\}$ is a smooth curve (Hilbert polynomial $20t - 50$: degree 20, genus 51).

- `computations/sage/beauville_ii.sage`: the rank-≤ 1 locus of the web is empty, and Beauville's condition (ii) holds at each of the ten rank-2 points, computed exactly over $\overline{\mathbf{Q}}$.
- `computations/sage/defect2.sage`: the ideal of the nodes is radical of degree 10, and its Hilbert function is $1, 4, 10, 10, \ldots$, so the nodes impose independent conditions on quadrics.
- `computations/sage/bF2.sage`: the toric resolution of $\frac{1}{5}(1,1,4)$ by iterated star subdivision has 7 rays and 9 smooth maximal cones; an independent construction of the fan gives the same result.
- `computations/sage/symm.sage`: auxiliary experiments with random invariant webs.

The homology bookkeeping of §5 is implemented twice, with identical results: `computations/homology/lam.p` together with `w10.py` tracks torsion profiles, and `computations/homology/indep.py` together with `m10_indep.py` tracks degreewise integral groups with Künneth. `computations/homology/x12.py` contains the higher-dimensional scans. The MILP of Appendix A is `computations/milp/lpsearch2.py`, run by `computations/milp/run_lp.py` and `computations/milp/necessity.py`.

Statement on the use of AI

The results in this manuscript were found, checked and written up by Claude (Anthropic) agents working under the direction of the author. The computations listed above were run by those agents. The manuscript has not undergone human peer review.

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